Savitribai Phule Pune University
S.Y.B.Sc(Computer Science) NEP - 2023
S.Y.B.Sc.(CS) Practical Examination MAR/APRIL – 2025 - 2026
Lab Course SEC-251-CS-P
Computer Networks

Duration: 3 Hours | Maximum Marks: 35

Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) What is the purpose of a modem in a network? Can we connect directly to the internet without it?

b) What is a MAC address and how is it different from an IP address?

c) What is a loopback IP address? What is its range?

d) Write the command to assign a static IP in Linux.

f) Which online tool used to check password strength.

g) How to Use Nmap to Scan for Open Ports.

Q2) **B** Write a C program to implement the data link layer framing methods such as character and bit stuffing. [20 marks]

OR

Q2) **B** Write a C program to check password is strong or weak. [20 marks]

OR

Q2) **B** **What is subnetting?** [20 marks]

Given the network address 192.168.10.0/25 (Class C):
Identify the number of network bits and host bits.

- **Find the total number of subnetworks in a given network.**
- **Calculate the total number of IP addresses available in each subnet.**
- **Determine the total number of usable host addresses in each subnet.**

Configure the following IP address on a PC and also on a Desktop end device in Cisco Packet Tracer. Verify the configuration using appropriate commands.

- **IP Address: 192.168.1.10**
- **Subnet Mask: 255.255.255.0**
- **Default Gateway: 192.168.1.1**

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) What is the functional difference between a switch and a router?

b) How many hosts are possible in a /26 subnet?

c) What are the differences between Cat5e, Cat6, and Cat7 Ethernet cables?

d) Write the steps to identify a phishing email.

f) What are the two analysis methods used for analyzing malware?

g) What is the use of the ping command in networking?

Q2) **B** Write a C program to Simple Hash Simulation. [20 marks]

OR

Q2) **B** Write a C program to implement framing using character count method in the data link. [20 marks]

OR

Q2) **B** **What is subnetting?**
Given the network address 192.168.10.0/26 (Class C):
Identify the number of network bits and host bits. [20 marks]

- Find the total number of subnetworks in a given network.
- Calculate the total number of IP addresses available in each subnet.
- Determine the total number of usable host addresses in each subnet.
.

To Configure the IP address on a PC (real system) and also on a Desktop end device (virtual PC) in Cisco Packet Tracer using the following network details. Verify the configuration using appropriate commands.

- IP Address: 192.168.2.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

Q3) Viva [05 marks]

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Computer Networks

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) What does the command `tracert` or `traceroute` do?

b) Differentiate between `ip addr` and `ifconfig` commands in Linux.

c) What does nslookup do?

d) List two advantages and two disadvantages of dynamic IP configuration.

f) What is the function of a switch in a computer network?

g) What is network topology? What are its applications?

Q2) **B** Write a C program to display Bus Topology. [20 marks]

/ Create and simulate a Bus Topology in Cisco Packet Tracer and verify communication between all nodes

OR

Q2) **B** Write a C program to implement even parity error detection. [20 marks]

OR

Q2) **B** [20 marks]

1. What is a CIDR value?
   List the CIDR values for Class A, Class B, and Class C public IP addresses.
2. Design and configure a network in Cisco Packet Tracer consisting of two LANs connected through a Router (2911). Assign IP addresses as given and verify connectivity between all PCs using the ping command.
   - **Network Details: LAN 1 (10.0.0.0/8)**

   PC1 → **10.0.0.1** PC2 → **10.0.0.2** PC3 → **10.0.0.3**
   Subnet Mask → **255.0.0.0**
   - **LAN 2 (192.168.1.0/24)**

   PC4 → **192.168.1.1** PC5 → **192.168.1.2** PC6 → **192.168.1.3**
   Subnet Mask → **255.255.255.0**
   - **Router (2911 Configuration)**

   Interface G0/0 → **10.0.0.4 / 255.0.0.0**
   Interface G0/1 → **192.168.1.4 / 255.255.255.0**

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) What is Cyber Attack? Write the types of Cyberattacks.

b) How can you reset or release an IP address obtained from DHCP?

c) Explain supernetting with an example combining IPv4 Class C networks.

d) How do you crack passwords with John the Ripper?

f) Explain the use of copy running-config startup-config.

g) What is a loopback IP address? What is its range?

Q2) **B** Write a C program that reads an email from a file and performs phishing pattern detection. [20 marks]

OR

Q2) **B** Write a C program that checks private IP ranges. [20 marks]

OR

Q2) **B** Create and configure a simple network in Cisco Packet Tracer by connecting two PCs through a switch. Assign the IP addresses 192.168.1.1 and 192.168.1.2 with subnet mask 255.255.255.0 to the PCs, and verify network connectivity using the ping command. [20 marks]

Explain the IP address classes (Class A, B, C, D, and E) and their IP address ranges.

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) What does the `ip link set eth0` up command do?

b) How to detect hidden Wi-Fi networks.

c) Write the command sequence to configure a router interface with IP.

d) What is the purpose of assigning a hostname to a switch?

f) What is the ethical hacking?

g) What does the show `running-config` command display?

Q2) **B** Write a C program to display a Ring Topology. / [20 marks]

Create and simulate a Ring Topology in Cisco Packet Tracer and verify communication between all nodes.

OR

Q2) **B** Write a program to implement odd parity error detection. [20 marks]

OR

Q2) **B** Test Ethernet LAN cables using an NSS-468A LAN Tester and verify network connectivity by checking Straight-Through or Cross-Over RJ-45 cables. [20 marks]

To create and configure a simple network in Cisco Packet Tracer by connecting two PCs through a switch. Assign the IP addresses 10.0.0.1 and 10.0.0.2 with subnet mask 255.0.0.0 to the PCs, and verify network connectivity using the ping command.

Q3) Viva [05 marks]

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Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) Identify the network device from a given physical sample or image and write down its purpose and ports.

b) List different types of network cables and their uses of Coaxial, Twisted Pair (STP & UTP),Fiber Optic

c) What is the role of no shutdown in router configuration?

d) Compare the output of ip addr and ifconfig

f) What command do you type to move from Router> to Router# mode?

g) How many hexadecimal digits are needed for a 64-bit WEP key? Give any 5 examples.

Q2) B Write a program to implement framing using character count method in the data link layer. [20 marks]

OR

Q2) B Write a program to display Star Topology. / [20 marks]
Create and configure a Star Topology in Cisco Packet Tracer using one switch (2960) and five PCs (PC1, PC2, PC3, PC4, PC5). Assign the following IP addresses to the PCs and verify network connectivity using the ping command.

PC	IP Address	Subnet Mask
PC1	192.168.1.1	255.255.255.0
PC2	192.168.1.2	255.255.255.0
PC3	192.168.1.3	255.255.255.0
PC4	192.168.1.4	255.255.255.0
PC5	192.168.1.5	255.255.255.0

OR

Q2) B To create and configure a network in Cisco Packet Tracer using two PCs and one router. Assign the following IP addresses, configure the router interfaces, and verify network connectivity using ipconfig, ping, and tracert commands. [20 marks]

Given Configuration:

- **PC1** → 192.168.1.2 / 255.255.255.0
- **PC2** → 192.168.2.2 / 255.255.255.0
- **Router G0/0** → 192.168.1.1 / 255.255.255.0
- **Router G0/1** → 192.168.2.1 / 255.255.255.

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) Explain the difference between `ping` and `traceroute`.

b) Explain steps to identify a phishing email.

c) What is the function of a switch in a computer network?

d) How many hexadecimal digits are needed for a 64-bit WEP key? Give any 5 examples.

f) What is the primary purpose of a banner motd? How can it serve both a legal and security function?

g) Create a dummy phishing email and explain indicators of phishing.

Q2) **B** Write a program to displays a Mesh Topology. [20 marks]

OR

Q2) **B** Write a program to configure hostname, enable password and encrypted secret password [20 marks]

OR

Q2) **B** **Create a Mesh Topology in Cisco Packet Tracer using four PCs. Assign the following IP addresses and verify connectivity between all nodes.** [20 marks]

- PC1 → 192.168.1.1 / 255.255.255.0
- PC2 → 192.168.1.2 / 255.255.255.0
- PC3 → 192.168.1.3 / 255.255.255.0
- PC4 → 192.168.1.4 / 255.255.255.0

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) Why is it important to configure an enable (EXEC mode) password on a switch?

b) Explain the role of VLAN1 in a switch's initial configuration.

c) Configure Static NAT on a router using Cisco Packet Tracer or GNS3. Assign a public IP to a private IP.

d) Configure Dynamic NAT with a pool of public IP addresses.

f) Define Hashing.

g) What are Cyber Security Policies.

Q2) **B** Write a program to verify the configuration using show running- config and ping commands. [20 marks]

OR

Q2) **B** Write a program to set up a dynamic routing protocols such as RIP, EIGRP or OSPF [20 marks]

OR

Q2) **B** To configure the initial settings of a Cisco Switch (Example 2960) in Cisco Packet Tracer by setting the hostname and console password using CLI commands. Verify the configuration using appropriate show commands. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) List and explain all five IP address classes with examples.

b) Explain the format and example of an IPv6 address

c) Differentiate between Static NAT, Dynamic NAT, and Port Address Translation (PAT)

d) Configure Dynamic NAT with a pool of public IP addresses.

f) Define Cyber Security.

g) Explain Ethical Hacking.

Q2) **B** Write a program to determine IP address class. [20 marks]

OR

Q2) **B** Write a program to implements Static Routing Simulation. [20 marks]

OR

Q2) **B** To create and configure a simple network in Cisco Packet Tracer by connecting two PCs through a switch. Assign the IP addresses **172.20.1.1** and **172.20.1.2** with subnet mask 255.255.0.0to the PCs, and verify network connectivity using the ping command. [20 marks]

Explain the IP address classes (Class A, B, C, D, and E) and their IP address ranges.

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) What are the differences between Cat5e, Cat6, and Cat7 Ethernet cables?

b) What is the purpose of a modem in a network? Can we connect directly to the internet without it?

c) Write the command to assign a static IP in Linux.

d) What is the use of the ping command in networking?

f) Differentiate between Static NAT, Dynamic NAT, and Port Address Translation (PAT).

g) Configure Static NAT on a router using Cisco Packet Tracer or GNS3. Assign a public IP to a private IP.

Q2) **B** Write a program to implement even parity error detection.. [20 marks]

OR

Q2) **B** Write a program to perform subnetting and supernetting calculations. [20 marks]

OR

Q2) **B** What is subnetting? [20 marks]
Given the network address 192.168.10.0/25 (Class C):
Identify the number of network bits and host bits.

- How many host bits are available?
- Determine the number of networks and number of hosts per network.
- Calculate the total number of assignable IP addresses in each network.

To create and configure a simple network in Cisco Packet Tracer by connecting two PCs through a switch. Assign the IP addresses 192.168.10.1 and 192.168.10.2 with subnet mask 255.255.255.0 to the PCs, and verify network connectivity using the ping command.

Q3) Viva [05 marks]

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Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) Compare and contrast the functionalities of a switch and a router.
b) What is the purpose of a modem in a network? Can we connect directly to the Internet without it?
c) List different types of network cables and their uses.
d) Define IPv4 and IPv6. Mention key differences between them.
f) What is subnetting? How many hosts are possible in a /26 subnet?
g) What is the purpose of assigning a hostname to a switch?

Q2) B Write a C program to implement framing using the Character Count Method. [20 marks]

OR

Q2) B Write a C program to simulate the Stop-and-Wait ARQ protocol. [20 marks]

OR

Q2) B Configure the initial settings of a Cisco switch by setting the VTY password, and privileged EXEC mode password using CLI commands in Cisco Packet Tracer, and verify the configuration using appropriate show commands. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) What is the function of a switch in a computer network?
b) Write the command to assign a static IP in Linux.
c) What does the `ip link set eth0 up` command do?
d) Explain the difference between `ping` and `traceroute`.
f) What does the `ip route show` command display?
g) Why should two connected systems be in the same subnet?

Q2) **B** Write a C program to display a Star Topology. / [20 marks]

OR

Q2) **B** Write a program to configure hostname, enable password and encrypted secret password [20 marks]

OR

Q2) **B** To create and configure a Star Topology in Cisco Packet Tracer using one switch (Example 2960) and five PCs (PC1, PC2, PC3, PC4, PC5). Assign the following IP addresses to the PCs and verify network connectivity using the ping command. [20 marks]

PC	IP Address	Subnet Mask
PC1	192.168.10.1	255.255.255.0
PC2	192.168.10.2	255.255.255.0
PC3	192.168.10.3	255.255.255.0
PC4	192.168.10.4	255.255.255.0
PC5	192.168.10.5	255.255.255.0

Q3) Viva [05 marks]

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Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) How do you verify if an interface received an IP dynamically?
b) Demonstrate assigning IPs and testing connectivity for 3 systems in a LAN.
c) Explain the role of VLAN1 in a switch's initial configuration.
d) What steps are required to configure the default gateway as 192.168.1.1.
f) What steps are required to configure Dynamic NAT with a pool of public IP addresses.
g) Use debug ip routing or debug ip rip to monitor routing events and updates.

Q2) B Write a program to displays a Hybrid Topology [20 marks]

OR

Q2) B Write a program to version that saves an Initial Switch Configuration into a text file [20 marks]

OR

Q2) B To create and simulate a Hybrid Topology in Cisco Packet Tracer by combining two different network topologies (e.g., Star and Bus or Star and Star). Assign appropriate IP addresses to all nodes and verify communication between them using the ping command [20 marks]

Topology Arrangement:

- **Star Network 1:**

PC1, PC2, PC3, PC4 → Switch1

- **Star Network 2:**

PC5, PC6, PC7, PC8 → Switch2

- Connect **Switch1 ↔ Switch2**

PC	IP Address	Subnet Mask
PC1	192.168.1.1	255.255.255.0
PC2	192.168.1.2	255.255.255.0
PC3	192.168.1.3	255.255.255.0
PC4	192.168.1.4	255.255.255.0
PC5	192.168.1.5	255.255.255.0
PC6	192.168.1.6	255.255.255.0
PC7	192.168.1.7	255.255.255.0
PC8	192.168.1.8	255.255.255.0

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) Configure PAT (NAT overload) to allow multiple internal hosts to access the internet using a single public IP.

b) Verify routing with `ping`, `traceroute`, and `show ip route` commands

c) Assign the privileged EXEC mode password as class.

d) Use show running-config to verify all the configurations.

f) List two advantages and two disadvantages of using dynamic IP configuration.

g) How do you verify if an interface received an IP dynamically?

Q2) **B** Write a program to implements Static Routing Simulation. [20 marks]

OR

Q2) **B** Write a program to program for LAN chat (client-server) using sockets [20 marks]

OR

Q2) **B** Configure a router by setting the hostname, enable password, and encrypted secret password, and configure console and VTY passwords for secure access. Also, set up a Linksys Wireless Router in Cisco Packet Tracer and configure the SSID for wireless network connectivity. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) What command do you type to move from `Router>` to `Router#` mode?

b) How many hexadecimal digits are needed for a 64-bit WEP key? Give any 5 examples.

c) Create a dummy phishing email and explain indicators of phishing.

d) Crack a zip file protected with a password using fcrackzip.

f) Compare the output of `ip addr` and `ifconfig`.

g) How can you reset or release an IP address obtained from DHCP?

Q2) **B** Write a program to Command-Line Phishing Simulation. [20 marks]

OR

Q2) **B** Write a program to displays a Mesh Topology. [20 marks]

OR

Q2) **B** **Create a Mesh Topology in Cisco Packet Tracer using four PCs. Assign the following IP addresses and verify connectivity between all nodes.** [20 marks]

- PC1 → 172.16.1.1 / 255.255.0.0
- PC2 → 172.16.1.2 / 255.255.0.0
- PC3 → 172.16.1.3 / 255.255.0.0
- PC4 → 172.16.1.4 / 255.255.0.0

Q3) Viva [05 marks]

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Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) Compare and contrast the functionalities of a switch and a router

b) What is a MAC address and how is it different from an IP address?

c) Perform subnetting on 192.168.1.0/26 and list all 4 subnet ranges with usable host IPs.

d) List and explain all five IP address classes with examples

f) How many hexadecimal digits are needed for a 64-bit WEP key? Give any 5 examples

g) What are two more secure wireless encryption standards that should be used instead of WEP in a real-world scenario?

Q2) B Write a C program to encrypt plain message to cipher message. [20 marks]

OR

Q2) B Write a C program to implement the any data link layer framing method. [20 marks]

OR

Q2) B Create and configure a Star Topology in Cisco Packet Tracer using one switch (Ex.2960) and five PCs (PC1, PC2, PC3, PC4, PC5). Assign the following IP addresses to the PCs and verify network connectivity using the ping command. [20 marks]

PC	IP Address	Subnet Mask
PC1	172.16.1.1	255.255.0.0
PC2	172.16.1.2	255.255.0.0
PC3	172.16.1.3	255.255.0.0
PC4	172.16.1.4	255.255.0.0
PC5	172.16.1.5	255.255.0.0

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) What is the primary purpose of a banner MOTD?

b) What is one benefit of using a simulator like Packet Tracer for learning networking, and what is one limitation compared to actual hardware?

c) Wha Differentiate between Static NAT, Dynamic NAT, and Port Address Translation (PAT).

d) List the commands used for Network Address Translation on Cisco's Packet Tracer.

f) Use `Netstat` command to view active connections.

g) What is the primary purpose of a banner MOTD?

Q2) **B** Write a C program for password security with Special Characters that includes special character checks along with uppercase, lowercase, digit, and length rules. [20 marks]

OR

Q2) **B** Write a C program to convert plain message to cipher message. [20 marks]

OR

Q2) **B** **What is subnetting?** [20 marks]

Given the network address 192.168.10.0/25 (Class C):
Identify the number of network bits and host bits.

- Find the total number of subnetworks in a given network.
- Calculate the total number of IP addresses available in each subnet.
- Determine the total number of usable host addresses in each subnet.

To create and configure a simple network in Cisco Packet Tracer by connecting two PCs through a switch. Assign the IP addresses 172.16.1.1 and 172.16.1.2 with subnet mask 255.255.0.0 to the PCs, and verify network connectivity using the ping command.

Q3) Viva [05 marks]

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Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) In Cisco's Packet tracer, which command used to change the prompt from *Router>* to *Router#* mode?

b) Write down the steps to connect a PC to the Linksys router's web-based configuration page and find the default IP address.

c) What is dynamic routing? Gives its features.

d) List the commands used for Network Address Translation on Cisco's Packet Tracer.

f) Explain steps to identify a phishing email.

g) Perform dictionary attack on SHA256 hash using Hashcat.

Q2) **B** Write a C program for Phishing Simulation [20 marks]

OR

Q2) **B** Write a C program that simulates a NAT system. [20 marks]

OR

Q2) **B** To create and simulate a Bus Topology using four PCs in Cisco Packet Tracer. Assign appropriate IP addresses to all nodes and verify communication between them using the ping command. [20 marks]

PC1 → 192.168.1.1 / 255.255.255.0

PC2 → 192.168.1.2 / 255.255.255.0

PC3 → 192.168.1.3 / 255.255.255.0

PC4 → 192.168.1.4 / 255.255.255.0

Q3) Viva [05 marks]

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Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) What is static routing? Gives its features.

b) List the commands used for Network Address Translation on Cisco's Packet Tracer.

c) Write an algorithm to Hash a password using SHA256.

d) Write the Linux commands used to detect hidden Wi-Fi networks.

f) Explain steps to identify a phishing email.

g) List types of Cyber Threats.

Q2) B Write a C program for Phishing Simulation. [20 marks]

OR

Q2) B Write a program to verify successful NAT translation using show NAT translation. [20 marks]

OR

Q2) B To create and configure a simple network in Cisco Packet Tracer using two PCs and one router. Assign the given IP addresses, configure the router interfaces, and verify network connectivity using ipconfig/ifconfig, ping, tracert, and nslookup commands. [20 marks]

Configuration:

- PC1 → 192.168.1.2 / 255.255.255.0
- PC2 → 192.168.2.2 / 255.255.255.0
- Router G0/0 → 192.168.1.1 / 255.255.255.0
- Router G0/1 → 192.168.2.1 / 255.255.255.0

Q3) Viva [05 marks]

Savitribai Phule Pune University
S.Y.B.Sc(Computer Science) NEP - 2023
S.Y.B.Sc.(CS) Practical Examination MAR/APRIL – 2025 - 2026
Lab Course SEC-251-CS-P
Computer Networks

Duration: 3 Hours Maximum Marks: 35

Q1) **A** **Write the answer of the following question (ANY 5)** [10 marks]

a) What is the primary purpose of a banner MOTD?

b) How many hexadecimal digits are needed for a 64-bit WEP key? Give any 5 examples

c) List types of Cyber Threats. Describe any two.

d) Differentiate between Static and Dynamic routing.

f) What is the function of a switch in a computer network?

g) Describe any two network topologies.

Q2) **B** Write a C program to represent Mesh Topology. [20 marks]

OR

Q2) **B** Write a C program to for Hash Simulation. [20 marks]

OR

Q2) **B** To create a Mesh Topology in Cisco Packet Tracer using four PCs. Assign the following IP addresses and verify connectivity between all nodes. [20 marks]

PC1 → 172.16.1.1 / 255.255.0.0

PC2 → 172.16.1.2 / 255.255.0.0

PC3 → 172.16.1.3 / 255.255.0.0

PC4 → 172.16.1.4 / 255.255.0.0

Q3) Viva [05 marks]